

Number Systems

Complete CAT Preparation Guide

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Topics

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Examples

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Practice Qs

~15%

CAT Weightage

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1.1 Types of Numbers

Every number in mathematics belongs to one or more categories. Understanding this hierarchy is the foundation of all Number System problems in CAT.

Type	Definition	Examples
Natural (N)	Counting numbers starting from 1	1, 2, 3, 4, 5 ...
Whole (W)	Natural numbers + zero	0, 1, 2, 3 ...
Integer (Z)	Whole numbers + negatives. No fractions.	..., -2, -1, 0, 1, 2 ...
Rational (Q)	Can be written as p/q ($q \neq 0$). Includes all terminating and repeating decimals.	$3/4$, -7, 0.25, $1/3$
Irrational	Cannot be written as p/q . Decimal never terminates or repeats.	$\sqrt{2}$, π , e
Real (R)	All rationals + all irrationals. Everything on the number line.	All of the above

CAT Trap: $\sqrt{4} = 2$, which is rational. Do not assume every square root is irrational. Always simplify first.

Even & Odd Number Rules

Operation	Result	Example
Even + Even	Even	$4 + 6 = 10$
Odd + Odd	Even	$3 + 5 = 8$
Even + Odd	Odd	$4 + 5 = 9$
Even $\times$ Even	Even	$4 \times 6 = 24$
Even $\times$ Odd	Even	$4 \times 3 = 12$
Odd $\times$ Odd	Odd	$3 \times 5 = 15$

Key insight: The **PRODUCT** of any set of integers is even if **AT LEAST ONE** of them is even. It is odd **ONLY** if **ALL** of them are odd.

1.2 Prime Numbers

A prime number has exactly two distinct factors: 1 and itself. Primes are the building blocks of all integers — every composite number is a unique product of primes (Fundamental Theorem of Arithmetic).

Primes up to 100 — Memorise These

2	3	5	7	11	13	17	19	23	29
31	37	41	43	47	53	59	61	67	71
73	79	83	89	97					

There are exactly **25 primes** below 100.

Critical Facts

- 1 is NOT a prime. It has only one factor (itself).
- 2 is the ONLY even prime number.
- All primes greater than 3 are of the form $6k \pm 1$ (but not all $6k \pm 1$ are prime — e.g. $25 = 6 \times 4 + 1$ is not prime).

How to Test if a Number N is Prime

Test divisibility by every prime up to $\sqrt{N}$. If none divides N, then N is prime.

Example: Is 127 prime?

$\sqrt{127} \approx 11.3 \rightarrow$ Test primes: 2, 3, 5, 7, 11

$127 \div 2 = 63.5$ (No)

$127 \div 3 = 42.3$ (No) $1+2+7 = 10$, not divisible by 3

$127 \div 5 = 25.4$ (No)

$127 \div 7 = 18.1$ (No)

$127 \div 11 = 11.5$ (No)

Conclusion: 127 is PRIME.

1.3 Divisibility Rules

These rules let you check divisibility without long division. In CAT, they save 20–30 seconds per question — which translates directly to marks.

Divisor	Rule	Example
2	Last digit is 0, 2, 4, 6, or 8	$4738 \div 2 \checkmark$ (last digit 8)
3	Sum of ALL digits divisible by 3	$4731 \rightarrow 4+7+3+1=15 \rightarrow 15 \div 3=5 \checkmark$
4	Last TWO digits form a number divisible by 4	$3748 \rightarrow 48 \div 4=12 \checkmark$
5	Last digit is 0 or 5	$3785 \checkmark$, $3783 \times$
6	Divisible by BOTH 2 AND 3	$312 \rightarrow$ even AND $3+1+2=6 \checkmark$
7	No simple rule — divide directly	—
8	Last THREE digits form a number divisible by 8	$54872 \rightarrow 872 \div 8=109 \checkmark$
9	Sum of ALL digits divisible by 9	$54819 \rightarrow 5+4+8+1+9=27 \checkmark$
10	Last digit is 0	$3490 \checkmark$
11	(Sum of odd-position digits) – (Sum of even-position digits) = 0 or multiple of 11	$83215 \rightarrow (8+2+5)-(3+1)=11 \checkmark$
12	Divisible by BOTH 3 AND 4	$144 \rightarrow 1+4+4=9 \checkmark$ AND $44 \div 4=11 \checkmark$
25	Last two digits are 00, 25, 50, or 75	$4375 \rightarrow 75 \checkmark$

Speed tip: For 6, 12, 15, 18 — always break them into two co-prime factors and test each separately. Divisible by 18 = divisible by 2 AND by 9.

Worked Example — Divisibility by 11

Example: Is 90321 divisible by 11?

Positions (left to right): 9(1st) 0(2nd) 3(3rd) 2(4th) 1(5th)

Sum of ODD positions (1st, 3rd, 5th) = $9 + 3 + 1 = 13$

Sum of EVEN positions (2nd, 4th) = $0 + 2 = 2$

Difference = $13 - 2 = 11 \rightarrow$ divisible by 11 $\checkmark$

1.4 HCF & LCM

Definitions

HCF (Highest Common Factor) — also called GCD. The largest number that divides all given numbers exactly.

LCM (Lowest Common Multiple) — The smallest number that is divisible by all given numbers.

$$\text{For any two numbers A and B : HCF(A, B) } \times \text{ LCM(A, B) = A } \times \text{ B}$$

Finding HCF — Prime Factorization

Express each number as a product of primes. HCF = product of **common** prime factors, each raised to its **lowest** power.

Example: HCF(72, 120)

$$72 = 2^3 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

Common factors: 2^3 and 3^1 (lowest powers)

$$\text{HCF} = 2^3 \times 3 = 8 \times 3 = \mathbf{24}$$

Finding LCM — Prime Factorization

LCM = product of **all** prime factors (from every number), each raised to its **highest** power.

Example: LCM(72, 120)

$$72 = 2^3 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

All distinct primes: 2, 3, 5 (highest powers: 2^3 , 3^2 , 5^1)

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 8 \times 9 \times 5 = \mathbf{360}$$

$$\text{Verify: HCF} \times \text{LCM} = 24 \times 360 = 8640 = 72 \times 120 \checkmark$$

CAT Application Types

Problem Type	Formula
Largest number dividing X and Y leaving same remainder r	$\text{HCF}(X - r, Y - r)$
Largest number dividing X, Y, Z leaving remainders a, b, c	$\text{HCF}(X - a, Y - b, Z - c)$
Smallest number divisible by X, Y, Z	$\text{LCM}(X, Y, Z)$

Smallest number that when divided by X, Y, Z leaves remainder r	$\text{LCM}(X, Y, Z) + r$
Largest number dividing X, Y, Z exactly	$\text{HCF}(X, Y, Z)$

CAT Trap: $\text{HCF} \times \text{LCM} = A \times B$ holds ONLY for exactly TWO numbers. For three or more numbers, this relationship does not apply directly. Use prime factorization instead.

1.5 Factors & Number of Factors

If $N = p^a \times q^b \times r^c \times \dots$ (prime factorization), then:

Property	Formula
Number of factors	$(a+1)(b+1)(c+1)\dots$
Sum of all factors	$[(p^{a+1}-1)/(p-1)] \times [(q^{b+1}-1)/(q-1)] \times \dots$
Number of EVEN factors	Total factors – Odd factors (or: $b \times (a+1)(c+1)\dots$ if 2^b is the even part)
Number of ODD factors	Product of (power+1) of all ODD prime factors only

Full example: $N = 360$

$$360 = 2^3 \times 3^2 \times 5^1$$

$$\text{Number of factors} = (3+1)(2+1)(1+1) = 4 \times 3 \times 2 = \mathbf{24}$$

$$\text{Sum of all factors} = (1+2+4+8)(1+3+9)(1+5) = 15 \times 13 \times 6 = \mathbf{1170}$$

$$\text{Number of ODD factors} = (2+1)(1+1) = 6 \text{ [ignore the } 2^3 \text{ part]}$$

$$\text{Number of EVEN factors} = 24 - 6 = \mathbf{18}$$

A number is a perfect square IF AND ONLY IF it has an ODD number of factors. Reason: factors pair up as $(d, N/d)$, but $\sqrt{N}$ pairs with itself — leaving one unpaired.

1.6 Unit Digit & Cyclicity

The unit digit of a product depends ONLY on the unit digits of the numbers being multiplied. This lets you ignore everything except the last digit.

Example: Unit digit of 37×43

Only care about: $7 \times 3 = 21 \rightarrow$ unit digit = **1**

Cyclicity of Powers — Memorise This Table

Base	Unit digits of powers	Cycle length
1	1, 1, 1, 1 ...	1
2	2, 4, 8, 6, 2, 4, 8, 6 ...	4
3	3, 9, 7, 1, 3, 9, 7, 1 ...	4
4	4, 6, 4, 6 ...	2
5	5, 5, 5, 5 ...	1
6	6, 6, 6, 6 ...	1
7	7, 9, 3, 1, 7, 9, 3, 1 ...	4
8	8, 4, 2, 6, 8, 4, 2, 6 ...	4
9	9, 1, 9, 1 ...	2
0	0, 0, 0, 0 ...	1

How to Use Cyclicity

Divide the exponent by the cycle length. The remainder tells you which position in the cycle to use.

Example 1: Unit digit of 7^{53}

Cycle of 7: [7, 9, 3, 1] — length 4

$53 \div 4 = 13$ remainder 1

Remainder 1 $\rightarrow$ 1st position in cycle $\rightarrow$ unit digit = **7**

Example 2: Unit digit of 2^{100}

Cycle of 2: [2, 4, 8, 6] — length 4

$100 \div 4 = 25$ remainder 0

Remainder 0 $\rightarrow$ use LAST position in cycle $\rightarrow$ unit digit = **6**

1.7 Remainders

Remainder problems are among the highest-frequency topics in CAT Number Systems. The key is knowing which technique to apply when.

Fundamental Property

$$(a + b) \bmod D = [(a \bmod D) + (b \bmod D)] \bmod D$$

$$(a \times b) \bmod D = [(a \bmod D) \times (b \bmod D)] \bmod D$$

Pattern / Cyclicity Method

Example: Remainder when 2^{25} is divided by 7

$$2^1 \bmod 7 = 2$$

$$2^2 \bmod 7 = 4$$

$$2^3 \bmod 7 = 8 \bmod 7 = 1 \leftarrow \text{cycle resets here}$$

Cycle: [2, 4, 1] — length 3

$$25 \div 3 = 8 \text{ remainder } 1$$

Remainder 1 $\rightarrow$ 1st position $\rightarrow$ answer = 2

Fermat's Little Theorem

If p is prime and $\gcd(a, p) = 1$, then:

$$a^{(p-1)} \equiv 1 \pmod{p}$$

Example: Remainder when 2^{100} is divided by 7

$p = 7$ (prime), $a = 2$, $\gcd(2, 7) = 1$ — conditions met

By Fermat: $2^6 \equiv 1 \pmod{7}$

$$100 = 6 \times 16 + 4$$

$$2^{100} = (2^6)^{16} \times 2^4 \equiv 1^{16} \times 16 \equiv 16 \pmod{7} = 2$$

Common mistake: Fermat's theorem requires p to be PRIME and $\gcd(a, p) = 1$. If a is a multiple of p , the remainder is 0, not 1.

1.8 Factorials & Trailing Zeros

$n! = 1 \times 2 \times 3 \times \dots \times n$. Two main CAT questions: (1) trailing zeros in $n!$, and (2) largest power of a prime dividing $n!$.

Trailing Zeros in $n!$

A trailing zero is created by a factor of $10 = 2 \times 5$. Since 2s always outnumber 5s in $n!$, we only count the 5s.

$$\text{Trailing zeros in } n! = \text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$$

Example: Trailing zeros in $100!$

$$\text{floor}(100/5) = 20$$

$$\text{floor}(100/25) = 4$$

$$\text{floor}(100/125) = 0$$

$$\text{Total trailing zeros} = 20 + 4 + 0 = \mathbf{24}$$

Largest Power of a Prime p Dividing $n!$

This is Legendre's Formula:

$$\text{Highest power of prime } p \text{ in } n! = \text{floor}(n/p) + \text{floor}(n/p^2) + \text{floor}(n/p^3) + \dots$$

Example: Largest power of 3 that divides $20!$

$$\text{floor}(20/3) = 6$$

$$\text{floor}(20/9) = 2$$

$$\text{floor}(20/27) = 0$$

$$\text{Answer} = 6 + 2 + 0 = \mathbf{8} \text{ (so } 3^8 \text{ divides } 20! \text{ but } 3^9 \text{ does not)}$$

The same formula applies to find trailing zeros — trailing zeros = highest power of 5 in $n!$ (since 2s always exceed 5s, 5 is the bottleneck).

★ Practice Questions — Module 1

Attempt all questions without a calculator. Show your working — method matters as much as the answer.

Q1*Warm-up*

Is 561 prime or composite? Show your divisibility check step by step.

Q2*Divisibility*

Find the smallest 4-digit number divisible by 12.

Q3*HCF / LCM*

The LCM of two numbers is 180 and their HCF is 12. If one number is 36, find the other.

Q4*Factors*

How many factors does 7200 have?

Q5*Unit digit*

Find the unit digit of $3^{73} + 7^{54}$.

Q6*Remainder*

Find the remainder when 5^{99} is divided by 6.

Q7*Factorials*

Find the number of trailing zeros in 200!

Q8

Highest power

What is the largest power of 2 that divides 50!?

Q9

CAT Level

A number when divided by 6 gives remainder 4. When the same number is divided by 18, what are the possible remainders?

Quick Reference — Module 1

Formula	Notes
Number of factors of $p^a q^b r^c = (a+1)(b+1)(c+1)$	Core formula
$HCF(A,B) \times LCM(A,B) = A \times B$	Only for exactly 2 numbers
Trailing zeros in $n! = \sum \text{floor}(n / 5^k)$	Stop when $5^k > n$
$a^{p-1} \equiv 1 \pmod p$, p prime, $\gcd(a,p)=1$	Fermat's Little Theorem
Primes: last digit cycle [1,3,7,9]	All primes > 5 end in 1, 3, 7, or 9
Perfect square $\leftrightarrow$ odd number of factors	Key property

End of Module 1 — Number Systems